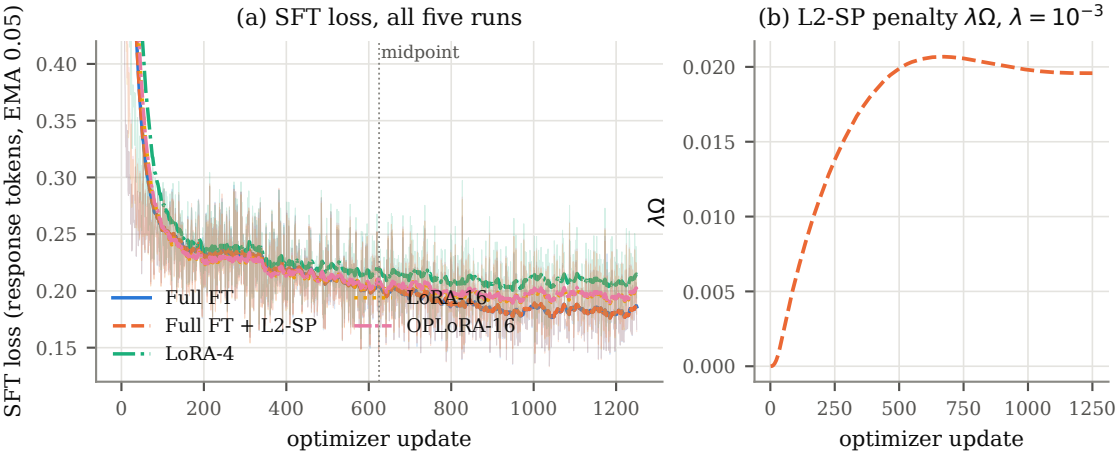

(a) SFT loss, all five runs
midpoint
SFT loss (response tokens, EMA 0.05)
0.40
0.35
0.30
0.25
0.20
0.15
0
200
400
600
800
1000
1200
optimizer update
Full FT
Full FT + L2-SP
LoRA-4
LoRA-16
OPLoRA-16
(b) L2-SP penalty λΩ, λ = 10⁻³
λΩ
0.020
0.015
0.010
0.005
0.000
0
250
500
750
1000
1250
optimizer update